

ATIVIDADE 8

• Pág. 99 - EXER. 4.16

$$4.16) \mu = 2,04 \text{ mm}; \text{ VAR} = 0,6084 \text{ mm}^2 \\ \sigma = 0,78 \text{ mm}$$

a) i) $P(X < 2,81) = 0,8389 = 83,89\%$

$$Z = \frac{X - \mu}{\sigma} = \frac{2,81 - 2,04}{0,78} = \frac{0,77}{0,78} = 0,99$$

ii) $P(X > 1,8) = P(Z > -0,31) = 1 - P(Z \leq -0,31) = 1 - 0,3783 = 0,6217 = 62,17\%$

$$Z = \frac{1,8 - 2,04}{0,78} = \frac{-0,24}{0,78} = -0,31$$

iii) $P(1,01 < X < 2,5) = P(X < 2,5) - P(X > 1,01) = 0,7224 - 0,0939 = 0,6290 = 62,9\%$

$$Z = \frac{1,01 - 2,04}{0,78} = \frac{-1,03}{0,78} = -1,32$$

$$Z = \frac{2,5 - 2,04}{0,78} = \frac{0,46}{0,78} = 0,59$$

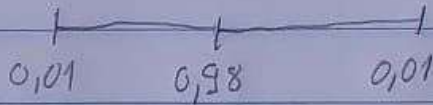
b) $P(2,2 < X < 3,8) = P(X < 3,8) - P(X > 2,2) = P(X < 3,8) - P(X < 2,2) =$
 $= 0,9881 - 0,5832 = 0,4049 \rightarrow 40,49\%$

$$Z = \frac{3,8 - 2,04}{0,78} = \frac{1,76}{0,78} = 2,26 \quad \bigg| \quad Z = \frac{2,2 - 2,04}{0,78} = \frac{0,16}{0,78} = 0,21$$

$$200 \cdot 0,4049 = 80,98 \text{ peças}$$

c) 98%:

$$c = 0,98$$



$$1 - 0,98 = 0,02$$

$$0,98 + 0,01 = 0,99 \rightarrow 99\%$$

$$\frac{0,02}{2} = 0,01$$

$$\downarrow$$
$$Z = 2,33$$

$$Z = \frac{x - \mu}{\sigma}$$

$$2,33 = \frac{x - 2,04}{0,78}$$

$$2,33 = \frac{x + 2,04}{0,78}$$

$$(2,33 \cdot 0,78) + 2,04 = x$$

$$2,04 \pm (2,33 \cdot 0,78)$$

$$x = 3,86$$

$$x = -0,22$$

$$[-0,22; 3,86]$$